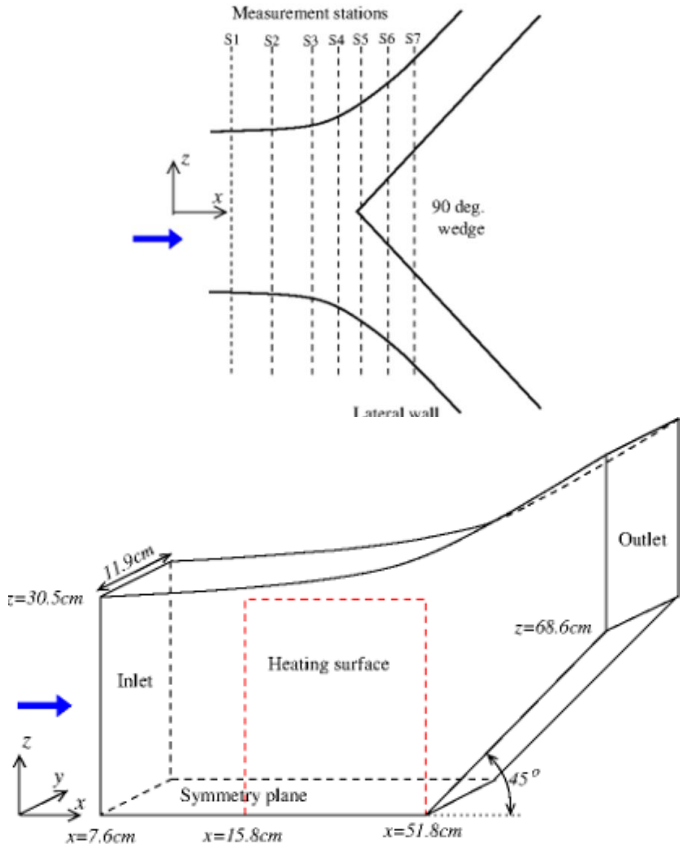
CFD Analysis of 3D Flow in a Wedge-Plate Junction Using Simcenter STAR-CCM+

Jeremiah Hasahatan Sitorus

Problem Description

- The wind-tunnel consists of two parallel planes separated by 11.9 cm with a development section for the preliminary 2D boundary layer and a test-section including the wedge and a heating surface.
- **Computational domain**: it is suggested to cover at least from $x = 7.6\ cm$ to $x = 121.9\ cm$.
- **Wedge**:
  - The vertex is located at $x = 51.8\ cm$ and $z = 0\ cm$.
  - Total angle of 90º and is positioned symmetrically.
  - Extends up to $x = 121.9\ cm$ and $z = \pm 68.6\ cm$.
- **Lateral walls**:
  - Satisfies fourth-order polynomial: $z = a + bx + cx^2 + dx^3 + ex^4$
  - $a = 30.004\ cm$,
  - $b = 0.018957\ cm$,
  - $c = -0.0036287\ cm^{-1}$,
  - $d = 0.00020397\ cm^{-2}$ and
  - $e = -0.00000119\ cm^{-3}$.



Wedge geometry. Plan view (top) and 3D view of half the geometry (bottom) [1]

[1] Source: "3D Flow in Wedge-Plate Junction," Case 002, ERCOFTAC Classic Collection, University of Manchester. Available at: http://cfd.mace.manchester.ac.uk/ercoftac/doku.php?id=cases:case002#img_cas2-geom

Problem Description

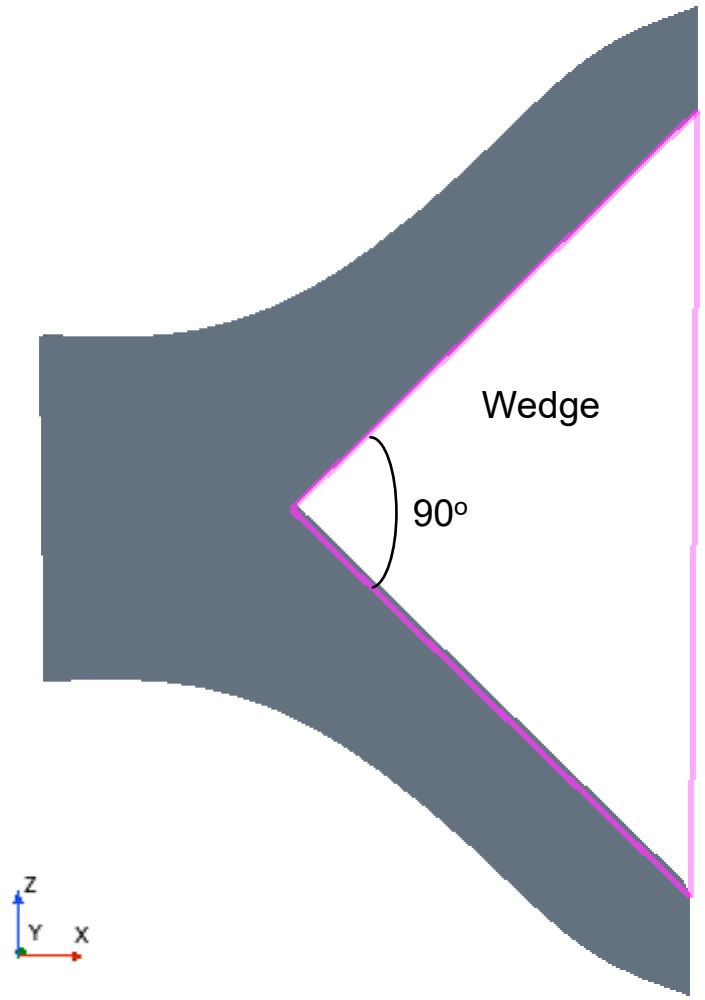
Flow Characteristics:

- The wedge creates a transverse pressure gradient.
- **Centerline**: the flow hits a strong adverse pressure gradient as it approaches the stagnation point in front of the wedge.
- **Off-centerline**: the pressure gradient starts adverse but then turns favorable as the flow accelerates around and past the wedge toward the exit.
- **Flow condition**:
  - Freestream velocity of $16.391\ m/s$,
  - kinematic viscosity of $1.5\times10^{-5}\ m^2/s$,
  - pressure of $770\ mmHg$.
- **Heating Surface:**
  - Dimension of $36\ cm \times 61\ cm$
  - Heat flux of around $630\ W/m^2$

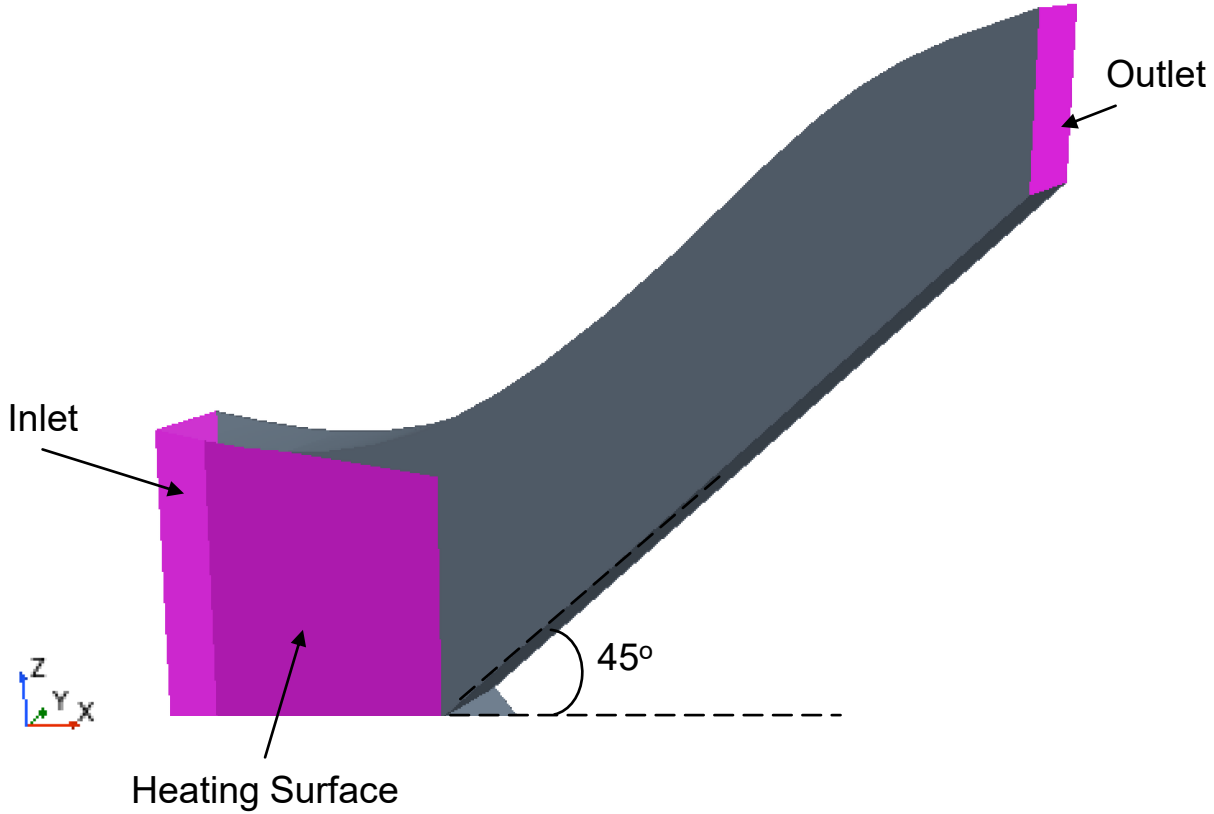
Measurement Probes:

S1	S2	S3	S4	S5	S6	S7
7.6 cm	22.9 cm	38.1 cm	45.7 cm	53.3 cm	61.0 cm	68.6 cm

Geometry and Mesh

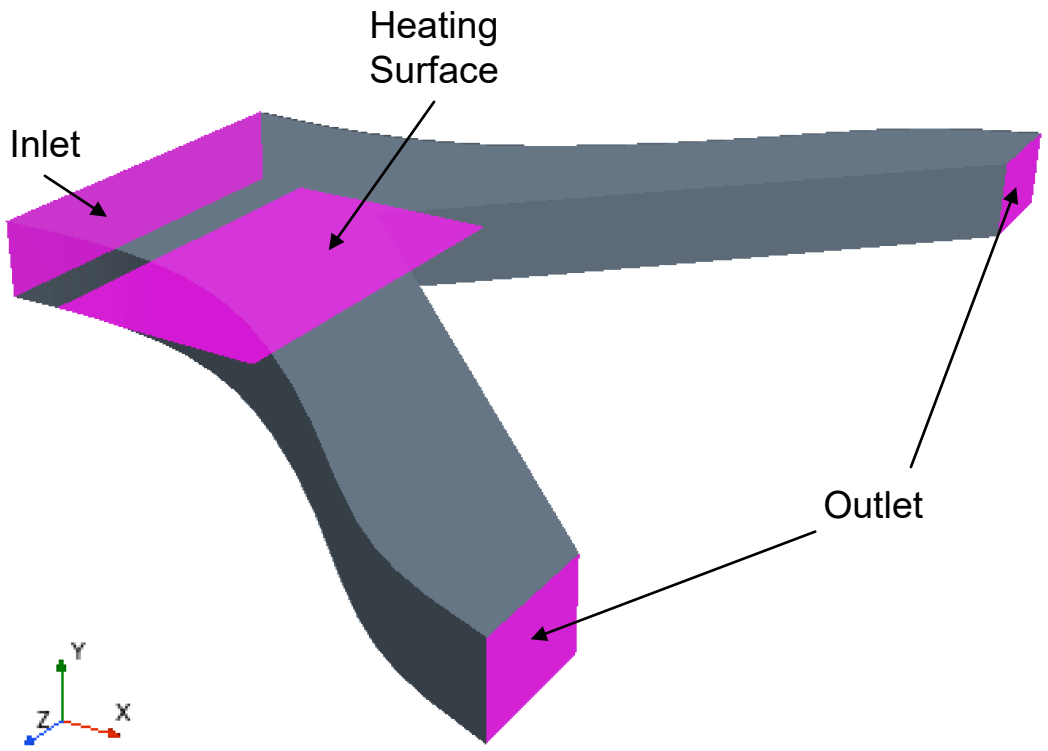


Plan view



3D view of half geometry

Geometry and Mesh



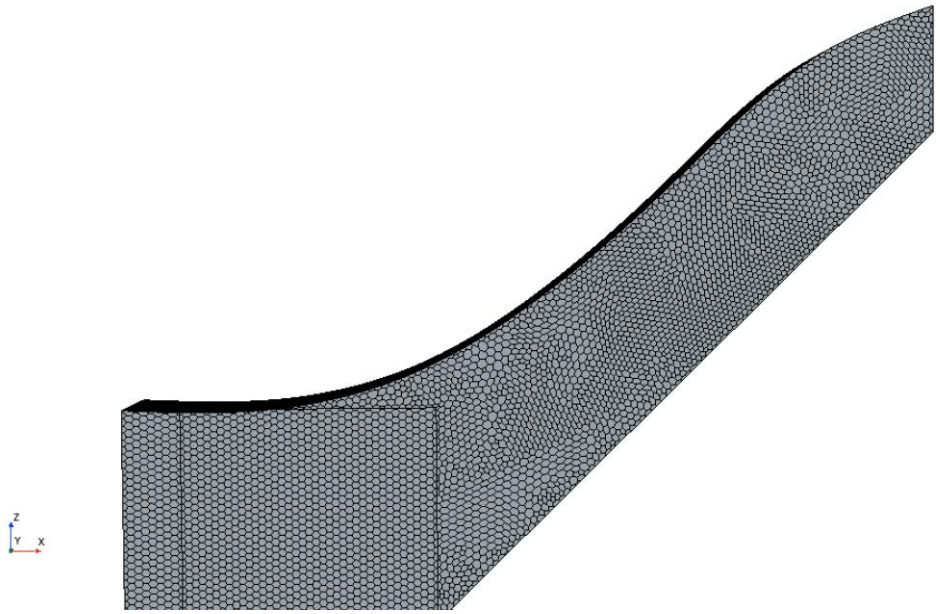
3D view of full geometry

Mesh

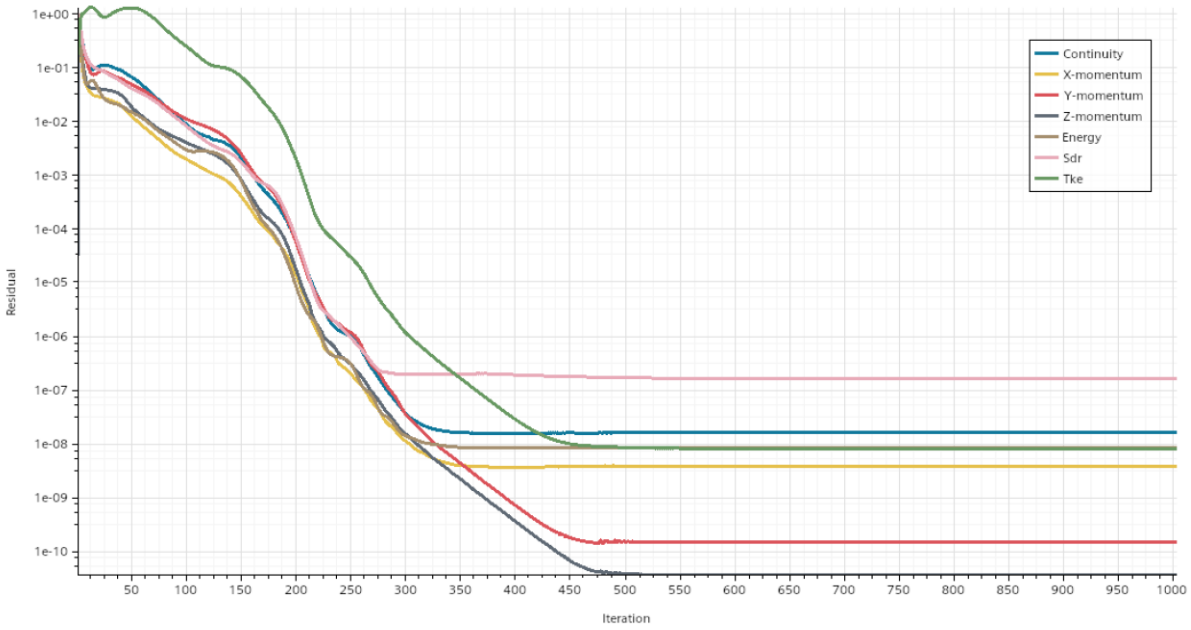
Automated Mesh

- Meshers:
  - Surface Remesher
  - Polyhedral Mesher
  - Prism Layer Mesher
- Base Size: 0.01 m
- No. of Prism Layers: 6
- Prism Layer Total Thickness: 0.01 m
- Prism Layer Stretching: 1.2

Geometry and Mesh



Mesh of the geometry



Convergence plot

CFD Setup

Physics

- All y+ Wall Treatment
- Gas
- Gradients
- Ideal Gas
- K-Omega Turbulence
- Reynolds-Averaged Navier-Stokes
- Segregated Flow
- Segregated Fluid Temperature
- SST (Menter) K-Omega
- Steady
- Three Dimensional
- Turbulent
- Wall Distance

Initial Condition

- Static Temperature = 300 K
- Turbulence Intensity = 0.01
- Turbulence Specification = Intensity + Viscosity Ratio
- Turbulence Velocity Scale = 1 m/s
- Turbulence Viscosity Ratio = 10
- Velocity = [0,0,0] m/s

Boundary Condition

Inlet

- Type: Velocity Inlet
- Temperature: 300 K
- Turbulence Intensity: 0.2%
- Turbulence Length Scale: 0.01 m
- Velocity: 16.391 m/s

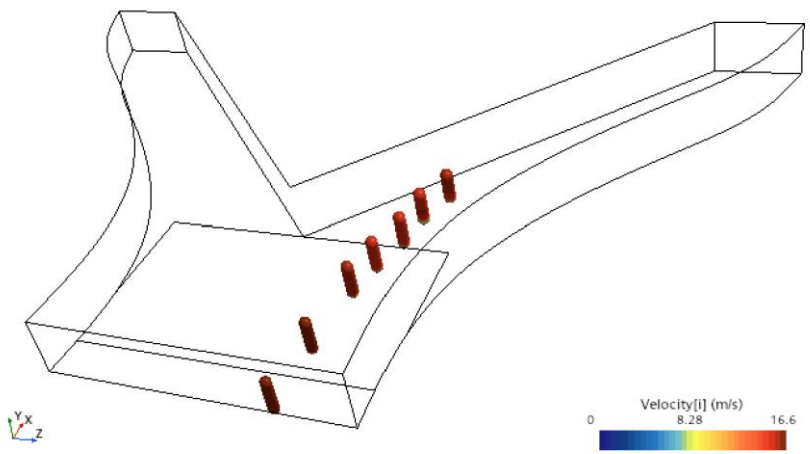
Outlet

- Type: Pressure Outlet
- Pressure: 0

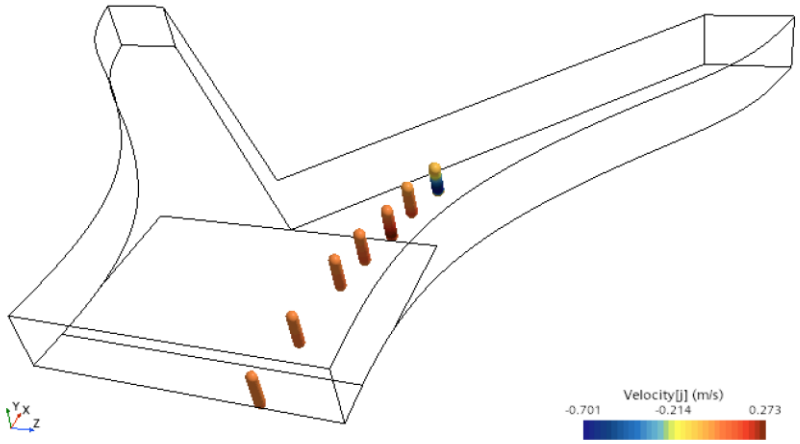
Heating Surface

- Type: Wall
- Thermal Specification: Heat Flux
- Value: 630 W/m²

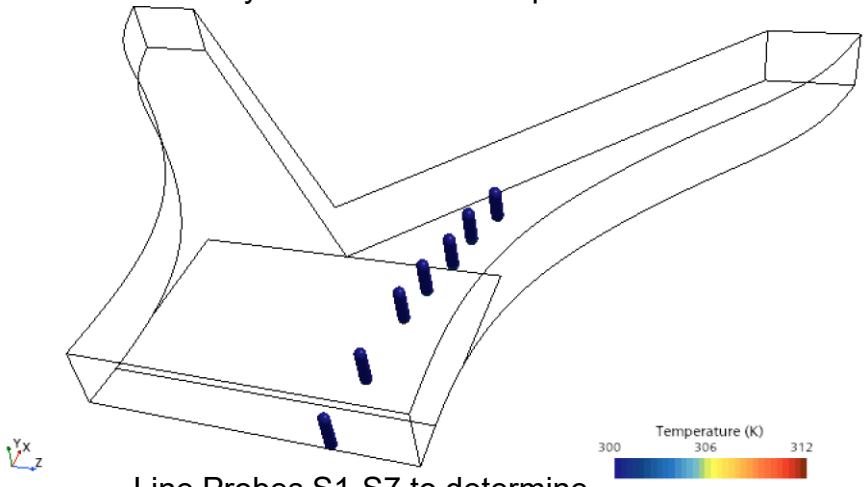
Flow Field Results



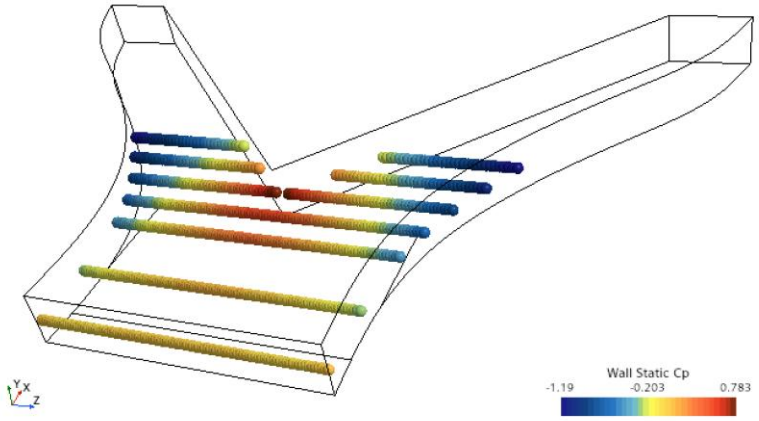
Line Probes S1-S7 to determine
Velocity in X-direction vs Y-position



Line Probes S1-S7 to determine
Velocity in Z-direction vs Y-position

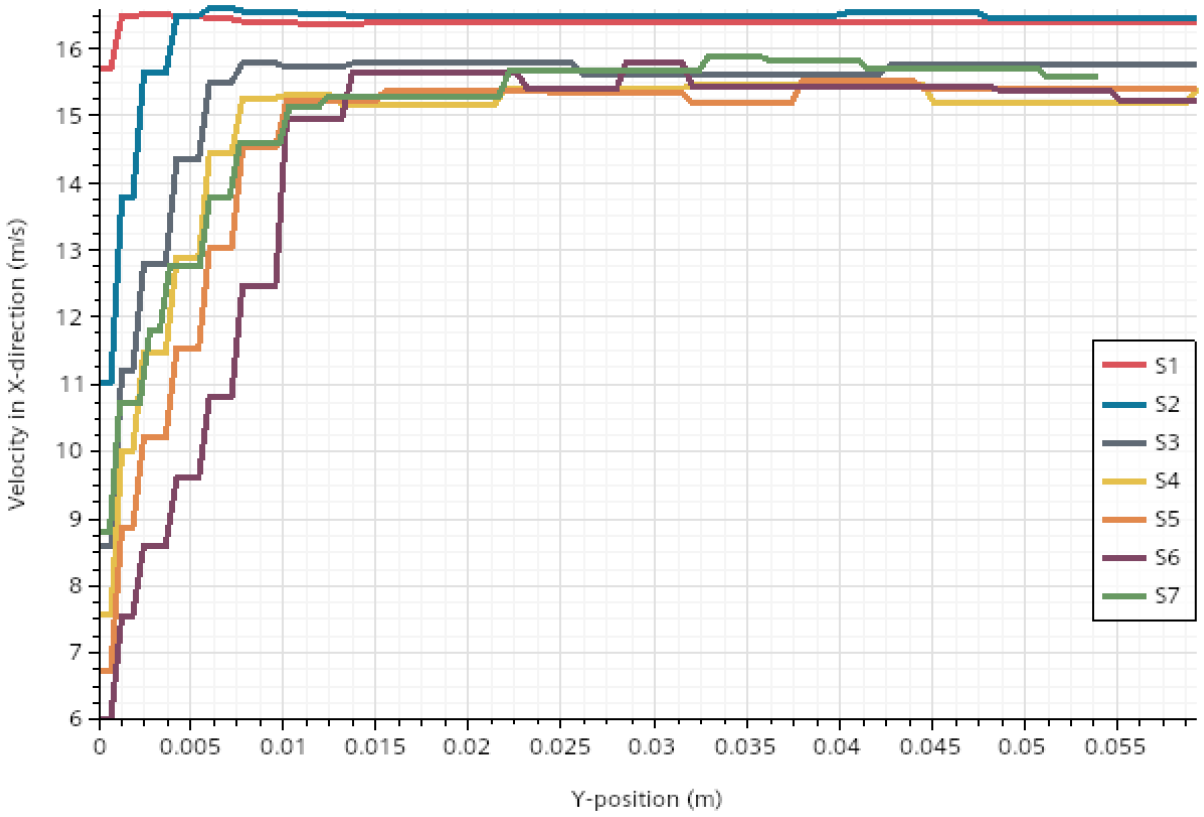


Line Probes S1-S7 to determine
Temperature vs Y-position

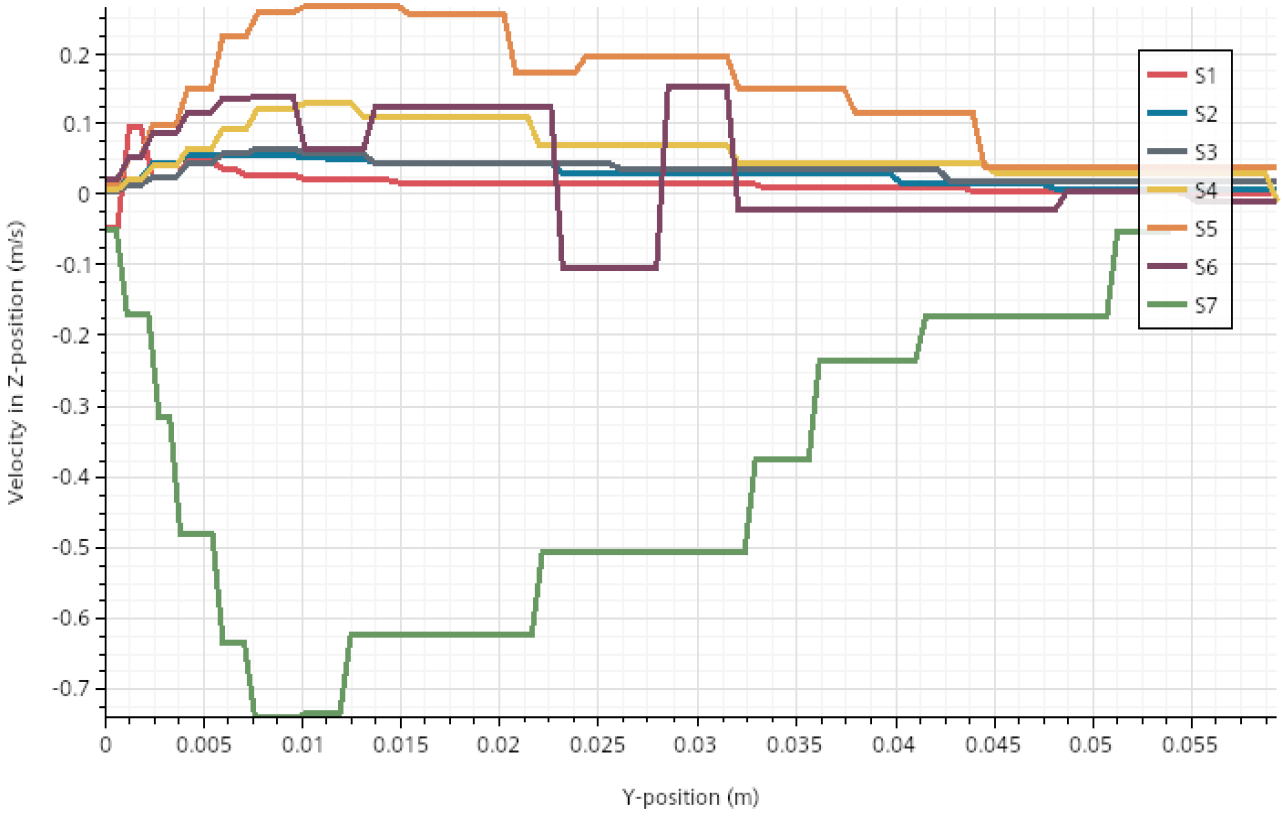


Line Probes S1-S7 to determine
Wall Static Cp vs Z-position

Flow Field Results

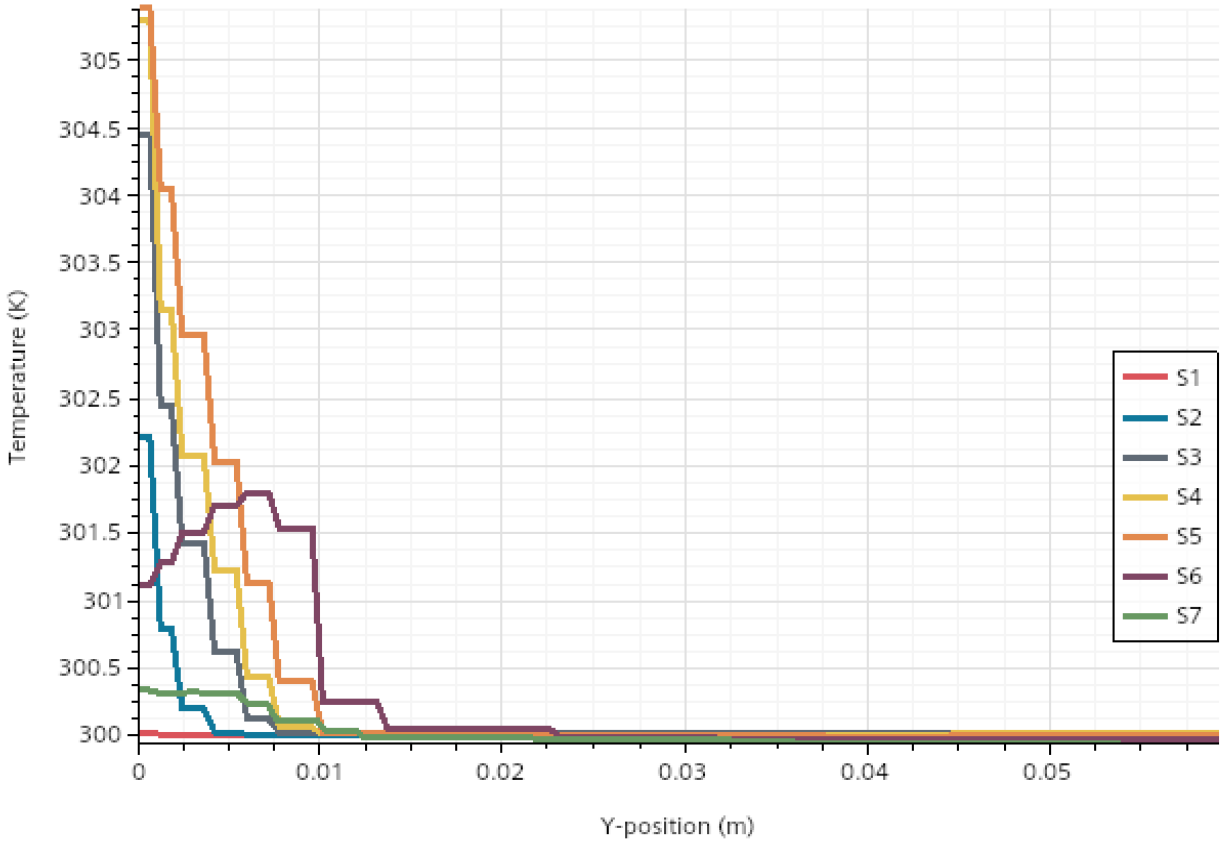


Velocity in X-direction vs Y-position in all measurement stations

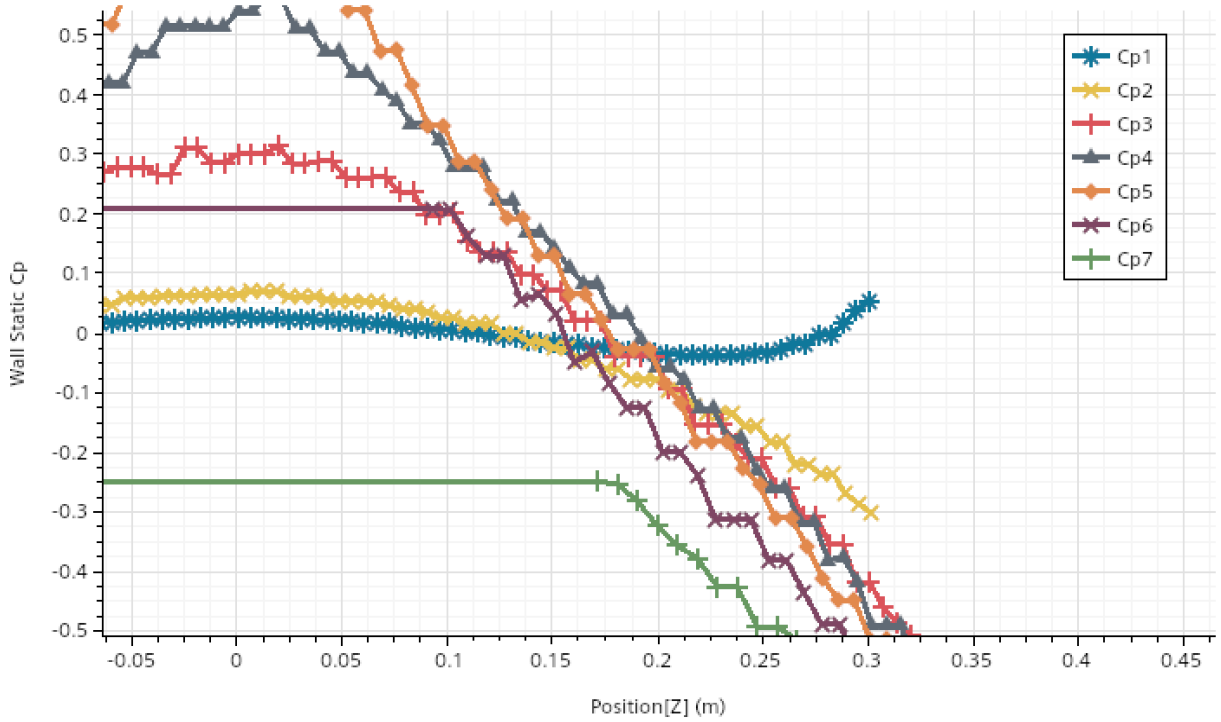


Velocity in Z-direction vs Y-position in all measurement stations

Flow Field Results



Temperature vs Y-position in all measurement stations



Wall Static Cp vs Z-direction in all measurement stations

Results Analysis

Streamwise Velocity

- Boundary layer develops along S1 to S7
- Velocity increases from near-wall low values and approaching free-stream
- Boundary layer thickness grows downstream

Spanwise Velocity

- Non-zero spanwise velocity appears downstream, especially S5 to S7
- Strong negative spanwise near the wall at S7, which indicates crossflow motion toward wedge centerline

Temperature

- High temperature near wall due to heating surface, then decrease to approximately 300 K away from wall
- Thermal boundary layer grows downstream

Pressure Coefficient

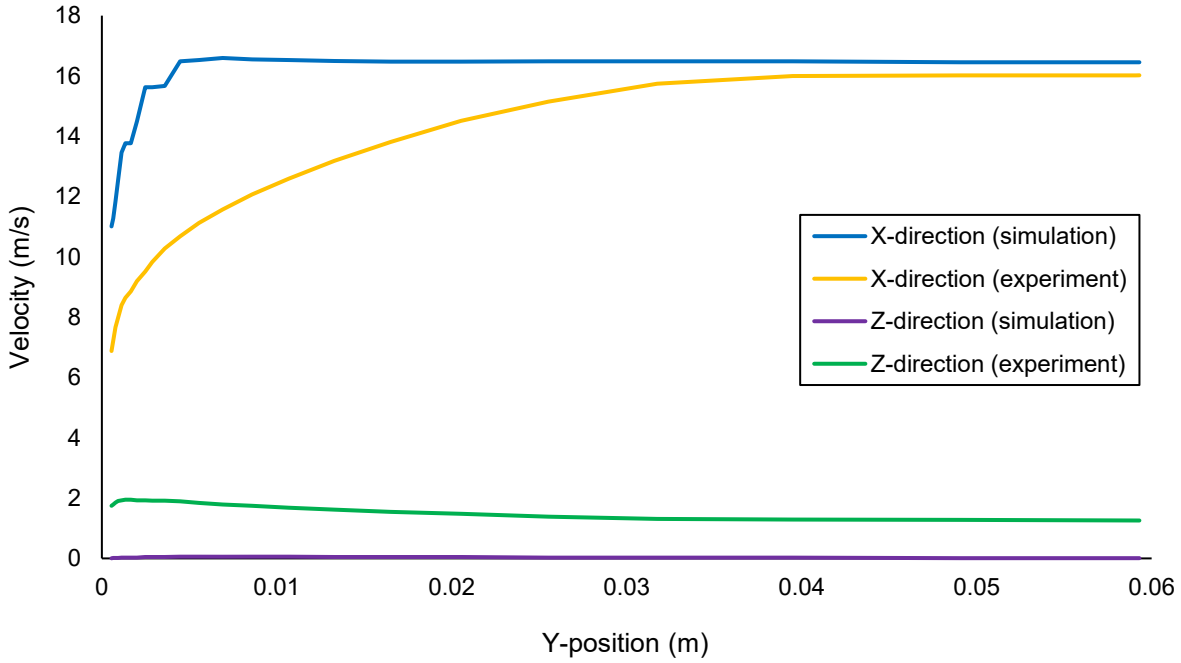
- High Cp near stagnation region (center)
- Cp decreases along Z-direction (flow acceleration)
- Strong pressure gradient across span

Comparison with Experiments

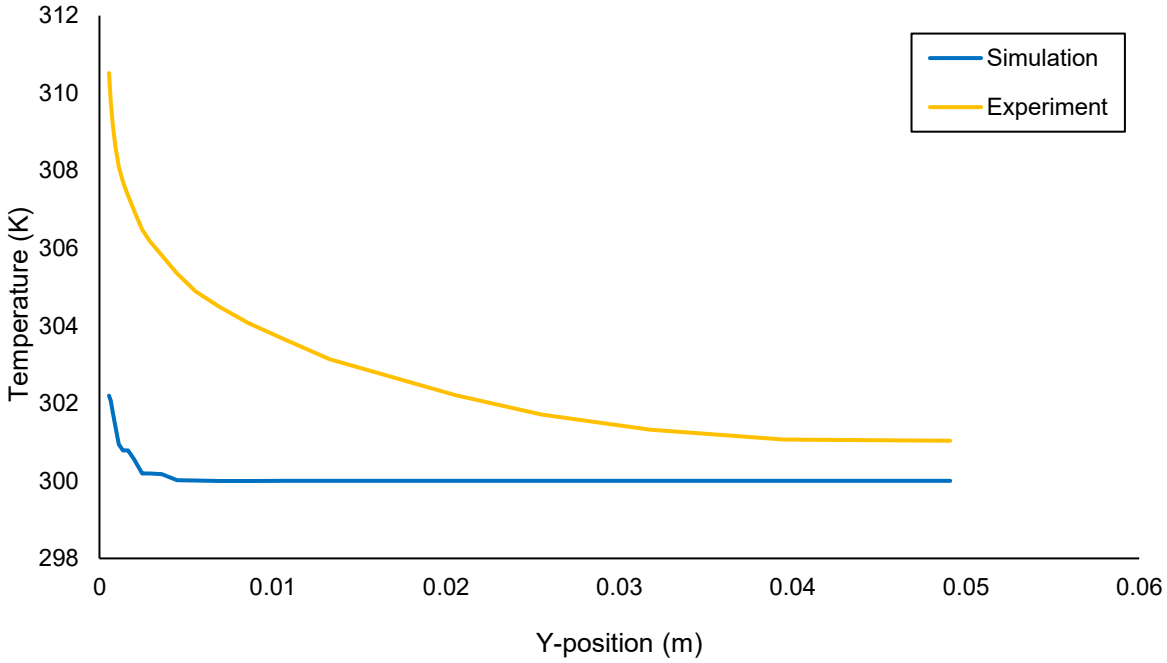
Measurement Station S2

X = 0.153 m
Z = 0.154 m

Velocity in X and Z-direction vs Y-position
for experiment and simulation



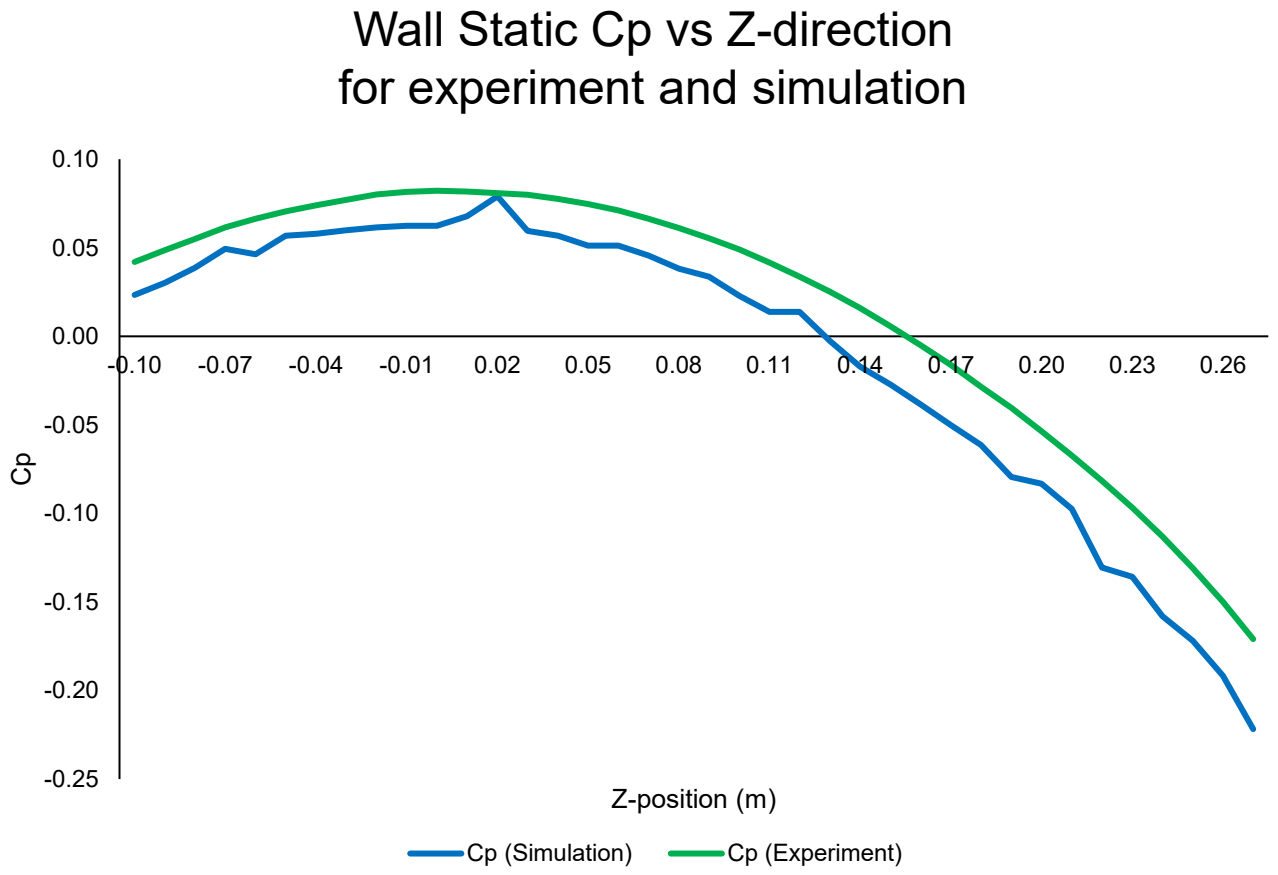
Temperature vs Y-position
for experiment and simulation



Comparison with Experiments

Measurement Station S2

X = 0.153 m
Y = 0.05 m

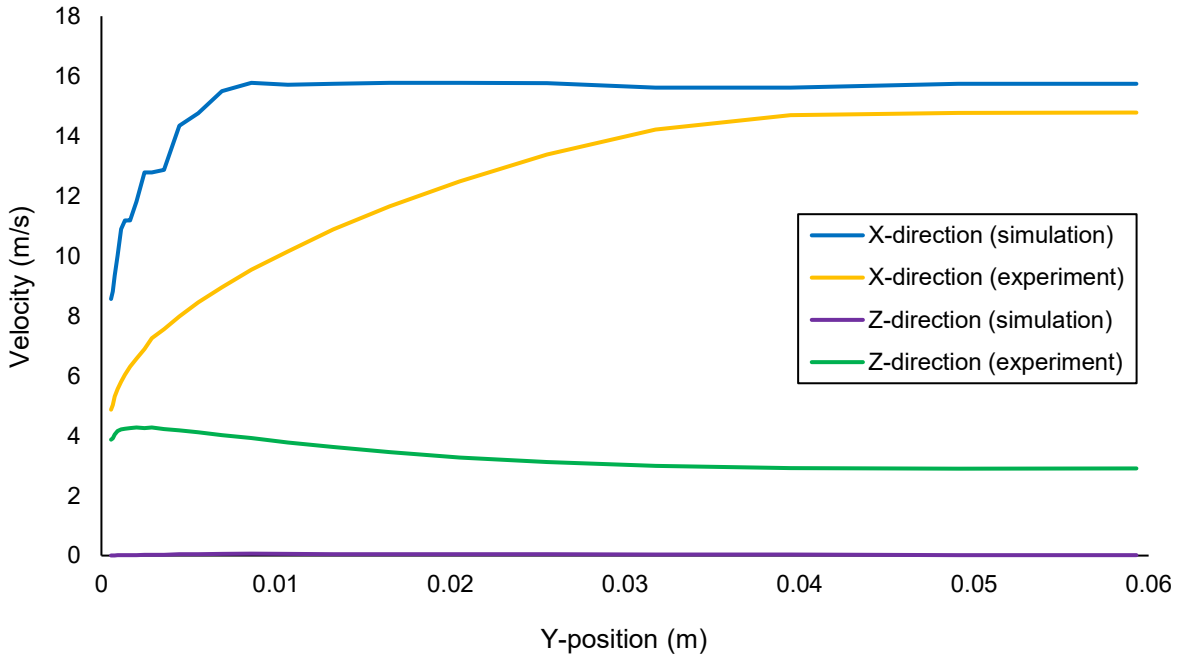


Comparison with Experiments

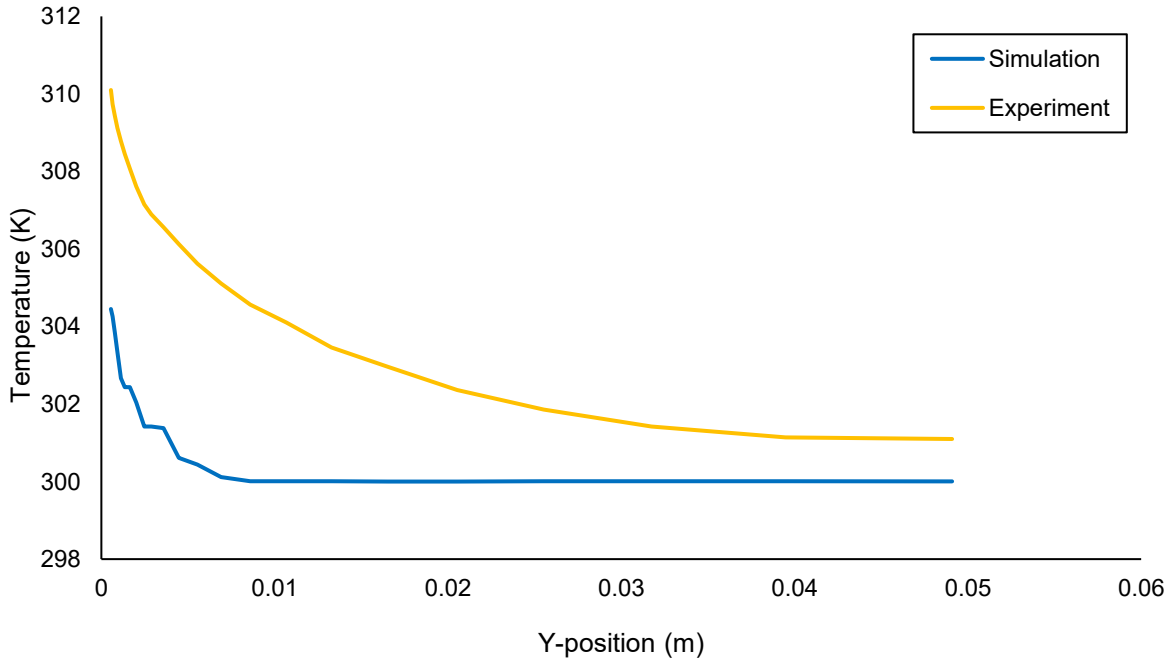
Measurement Station S3

X = 0.305 m
Z = 0.167 m

Velocity in X and Z-direction vs Y-position
for experiment and simulation



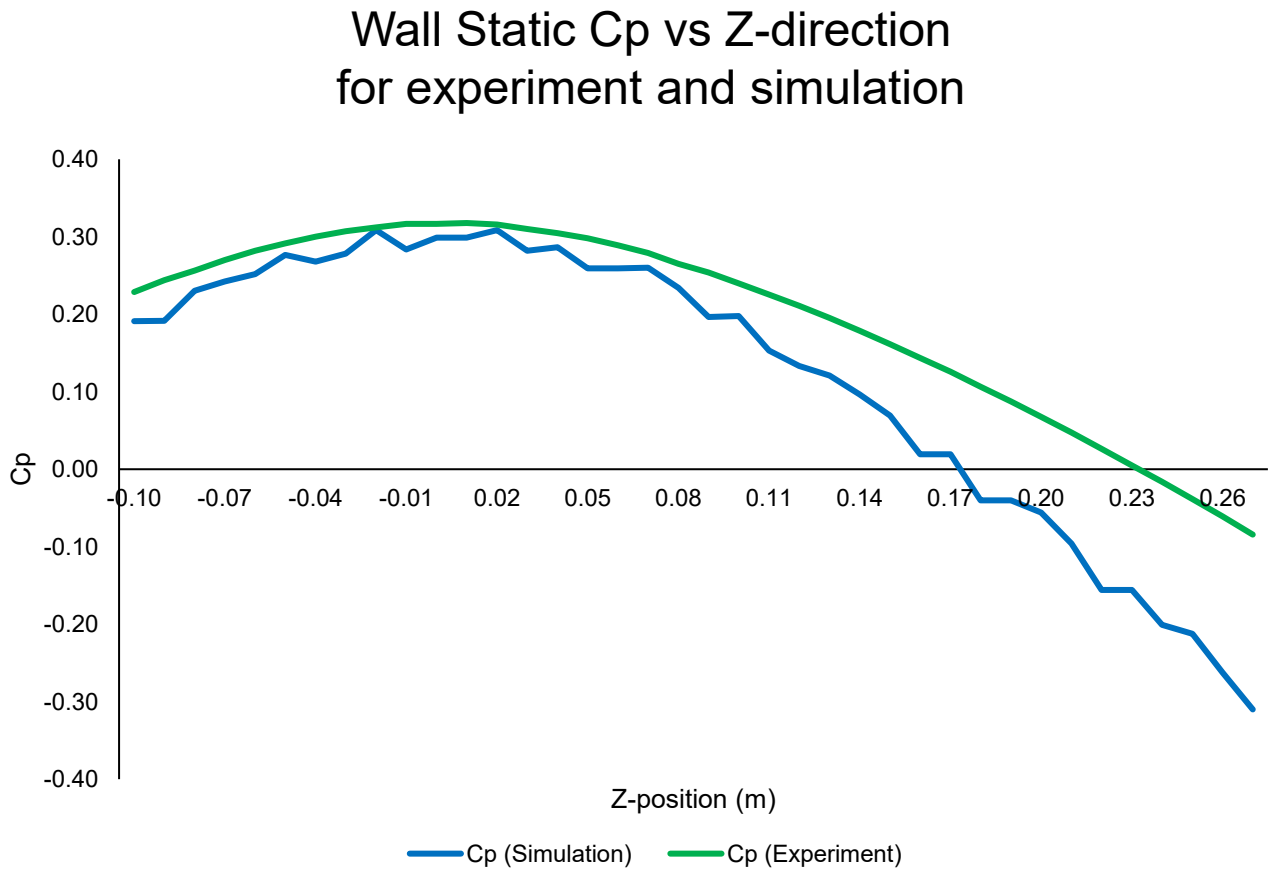
Temperature vs Y-position
for experiment and simulation



Comparison with Experiments

Measurement Station S3

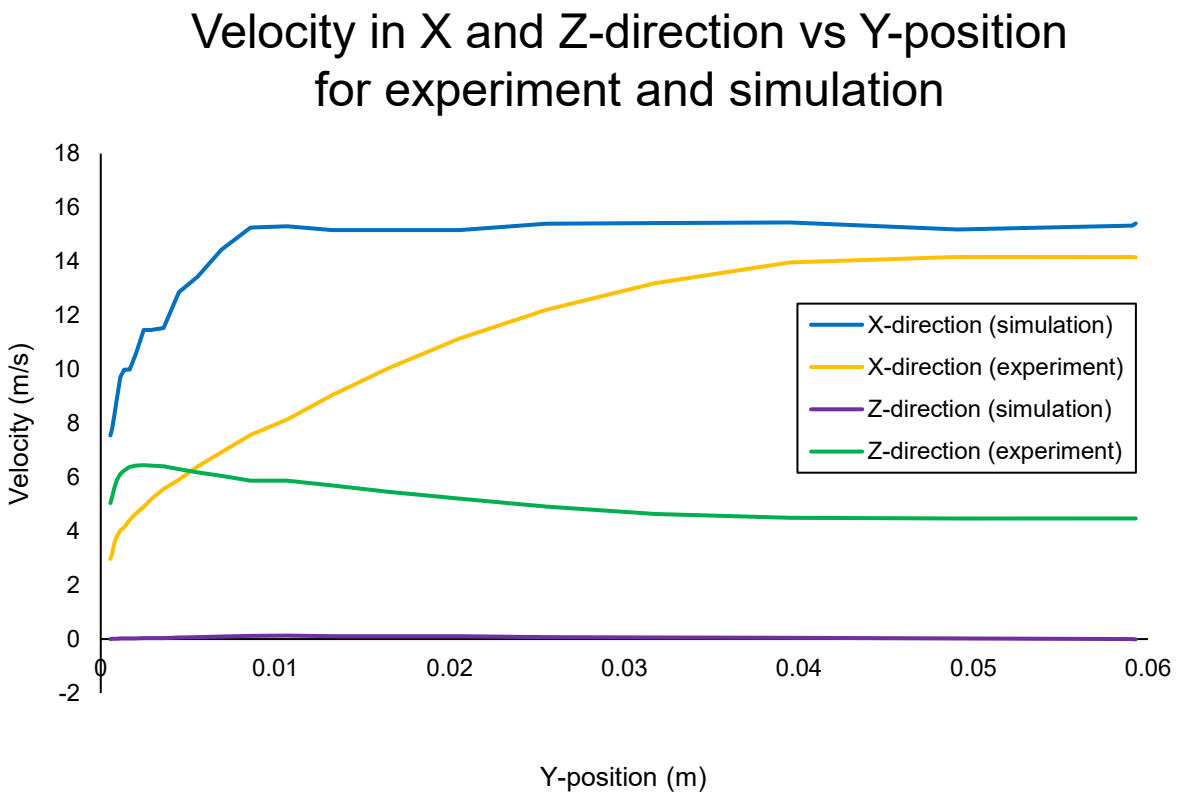
X = 0.305 m
Y = 0.05 m

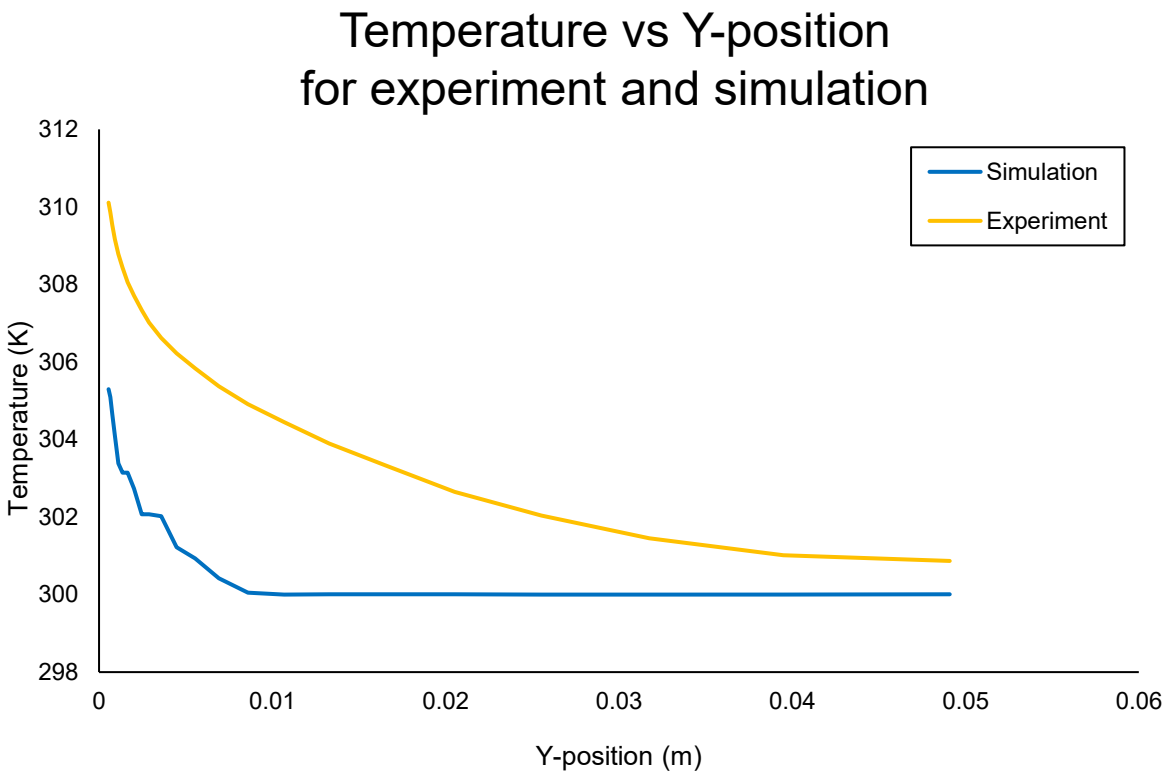


Comparison with Experiments

Measurement Station S4

X = 0.381 m
Z = 0.184 m

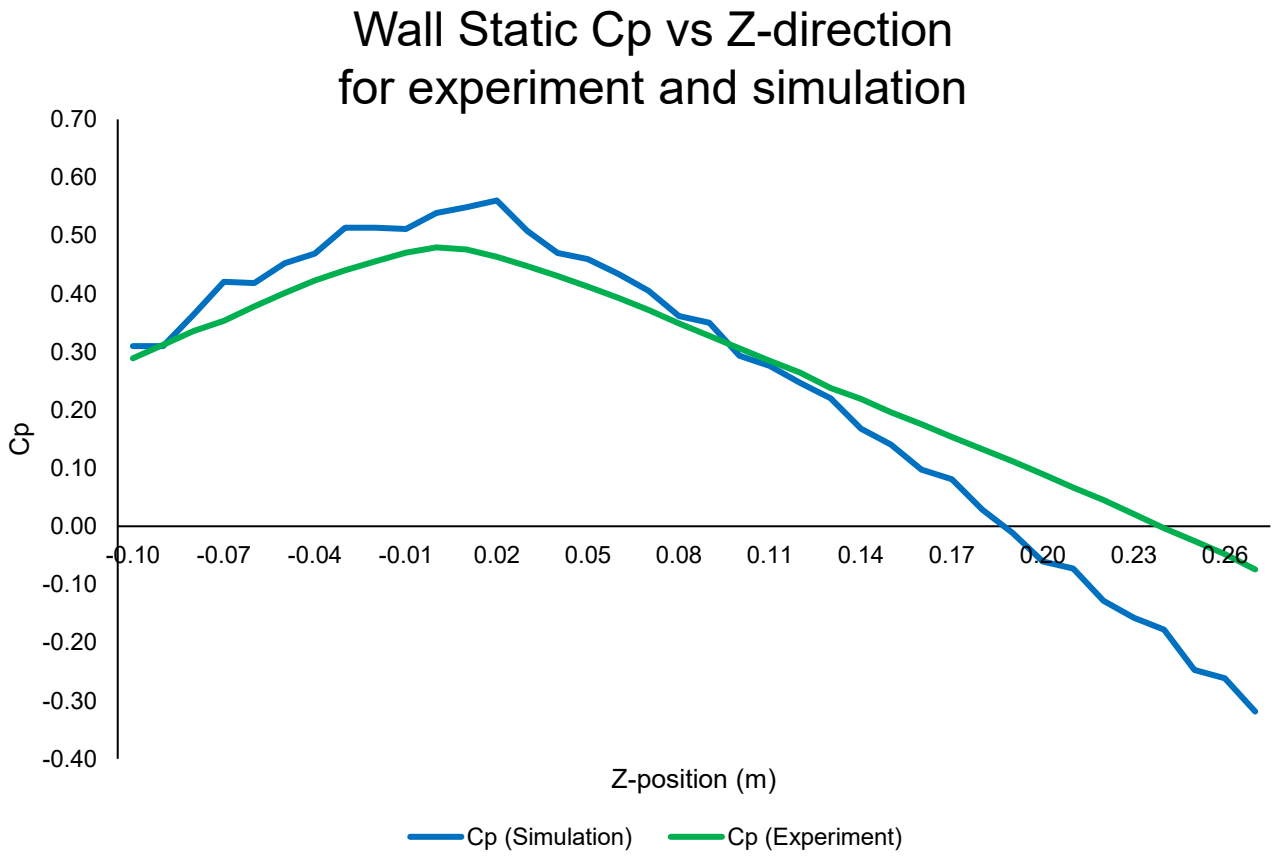




Comparison with Experiments

Measurement Station S4

X = 0.381 m
Y = 0.05 m



Comparison

Streamwise Velocity

- Simulation reach freestream faster than experiment
- Boundary layer is thinner in simulation

Spanwise Velocity

- Simulation do not capture the crossflow induced by transverse pressure gradient
- Possibly affected by mesh quality and physics assigned to the model

Temperature

- Simulation and experiment have high temperature near wall and decrease towards freestream, but the temperature decrease faster in simulation

Wall Static Cp

- Simulation capture similar pressure distribution pattern and peak locations as the experiment, but has faster pressure drop